

Problem G. Distância

Time Limit 4000 ms

Mem Limit 262144 kB

Suppose you have two strings S and t , and their length is equal. You may perform the following operation any number of times: choose two different characters C_1 and C_2 , and replace every occurrence of C_1 in both strings with C_2 . Let's denote the *distance* between strings S and t as the minimum number of operations required to make these strings equal. For example, if S is `abcd` and t is `ddcb`, the *distance* between them is 2 — we may replace every occurrence of `a` with `b`, so S becomes `bbcd`, and then we may replace every occurrence of `b` with `d`, so both strings become `ddcd`.

You are given two strings S and T . For every substring of S consisting of $|T|$ characters you have to determine the *distance* between this substring and T .

Input

The first line contains the string S , and the second — the string T ($1 \leq |T| \leq |S| \leq 125000$). Both strings consist of lowercase Latin letters from `a` to `f`.

Output

Print $|S| - |T| + 1$ integers. The i -th of these integers must be equal to the *distance* between the substring of S beginning at i -th index with length $|T|$ and the string T .

Examples

Input	Output
<code>abcdefa</code> <code>ddcb</code>	<code>2 3 3 3</code>